
Locality-Aware Adaptive Variational Quantum Dynamics on Fixed Qubit Topologies

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Abstract

Adaptive variational quantum dynamics simulation (AVQDS) compresses real-time evolution into shallow circuits by growing a pseudo-Trotter ansatz under McLachlan’s variational principle, but its growth procedure places two-qubit Pauli rotations between arbitrary qubit pairs, whereas near-term processors connect only nearest neighbors. We introduce TA-AVQDS (topology-adaptive AVQDS), which makes the ansatz-growth rule itself topology aware while leaving the variational equation of motion untouched. A persistent logical-to-physical qubit assignment is maintained and updated only by minimal SWAP migrations that are chosen—among all shortest-path realizations—to maximize the number of operator-pool pairs that remain nearest-neighbor; a pool operator is admitted without routing whenever its projected McLachlan-distance improvement is within a tolerance τ of the best unrestricted candidate, exploiting the near-degeneracy of candidate distances that is generic in adaptive growth. On the two eight-qubit benchmarks of Yao *et al.* [PRX Quantum 2, 030307 (2021)]—mixed-field Ising quenches and Lieb-Schultz-Mattis ramps—evaluated on a 2×4 square lattice and an 8-qubit heavy-hex subgraph, TA-AVQDS preserves the unconstrained fidelities with zero routing overhead on the square lattice, whose Hamiltonian cycle embeds both interaction graphs, and a mean CNOT overhead of $1.63\times$ on the heavy-hex subgraph, which admits no Hamiltonian path, compared with $2.48\times$ for post-hoc shortest-path SWAP routing of the same unconstrained circuit under the same provably optimal embedding. Ablations isolate the two mechanisms (permutation persistence and locality-aware migration with τ -admission), the pool-restriction limit without migration fails exactly where topology theory predicts, and all circuits are audited gate-by-gate against the topology edge sets. The approach generalizes to a linear-chain topology and adds only a small classical overhead to the growth loop.

1 Introduction

Simulating the real-time evolution of quantum many-body systems, $|\psi(t)\rangle = e^{-i\hat{H}t}|\psi(0)\rangle$, is a canonical application of quantum computers [1, 2, 3, 4]. On noisy intermediate-scale quantum (NISQ) devices [6], however, executable circuits are short, and first-order product-formula (Trotter) simulation [5, 2] requires depths that grow linearly in time, which rapidly exhausts the device budget for interacting spin systems [7, 8, 9]—the state’s complexity grows with time even at finite size [53, 50, 51], and the dynamics exhibits structure such as dynamical quantum phase transitions [49, 48] that accurate simulations must capture. Variational quantum dynamics (VQD) [10, 11, 12, 13, 14], an outgrowth of the variational eigensolver framework [46, 47, 34, 55, 56], mitigates this by evolving the parameters of a fixed ansatz circuit along McLachlan’s variational principle [15], but a fixed ansatz cannot anticipate the regions of Hilbert space that the wavefunction visits during a quench, and the simulation accuracy degrades accordingly.

Adaptive AVQDS [16] (see also Refs. [17, 18, 52]) removes this bottleneck by growing the ansatz during the simulation: at each time step the circuit is expanded, gates drawn from the Pauli-operator pool of the Hamiltonian, until the McLachlan distance L^2 —a hardware-measurable residual that bounds the local simulation error—falls below a threshold. ADAPT-style growth [19, 20] yields circuits with up to two orders of magnitude fewer CNOTs than Trotter circuits of matched accuracy [16], and the framework connects to operator pools with symmetry reduction [62, 63] and to imaginary-time variants [22, 23, 24].

The original formulation, however, is topology agnostic: the pool contains two-site operators between arbitrary qubit pairs, and its CNOT accounting assumes all-to-all connectivity (two CNOTs per two-qubit Pauli rotation) [16, 44]. Present superconducting processors are sparse graphs—IBM’s heavy-hex lattices [25] and Google-style square lattices [26] have vertex degree two to three—so a two-qubit gate between non-adjacent qubits must be preceded by SWAP gates that move the logical qubits together. The standard remedy is a post-hoc routing pass: Qiskit’s transpiler [33], t|ket> [28], and the academic routing literature [27, 29, 30, 31, 32] provide increasingly sophisticated SWAP-insertion heuristics, all of which operate on a *finished* circuit. For adaptively grown circuits this is wasteful in a specific, exploitable way: the growth loop continuously evaluates many candidate operators and their projected McLachlan distances, and these candidate distances are frequently near-degenerate across the pool—a fact already noted in Ref. [16] as the reason Hamiltonian pools outperform expanded pools. Near-degeneracy means the dynamics usually does not care which of several operators is appended, but the *topology* does, because different operators act on different qubit pairs. Conversely, when routing is unavoidable, the routing decision should account for which future operators will be needed—information that is available inside the growth loop but not to a post-hoc compiler.

This paper turns that observation into an algorithm. We propose *topology-adaptive AVQDS* (TA-AVQDS), which modifies exactly one ingredient of AVQDS: the operator-admission rule of the growth loop. The method (Sec. 3) maintains a logical-to-physical assignment π alongside the growing circuit. A pool operator is *admissible* if its operands are adjacent under π . At each growth event, the best admissible operator is admitted if its projected McLachlan-distance removal is within a tolerance τ of the best unrestricted candidate; otherwise the best unrestricted operator is admitted after a minimal chain of SWAP gates—at most $d(\pi(i), \pi(j)) - 1$ for operands at graph distance d —migrates its operands together, where the migration is chosen among all shortest paths and split points to maximize the number of pool pairs that are adjacent after the migration. The assignment persists, so each migration’s cost is amortized over subsequent gates. Crucially, a SWAP appended between the existing circuit and a new gate acts as a pure relabeling of the qubits carrying the logical sites: the variational state, the McLachlan equation of motion, the Tikhonov-regularized linear solve, and the adaptive step size are untouched by the routing layer. Topology awareness enters *only* through which operators the growth loop appends. With $\tau = 0$ and a complete graph the method reduces exactly to unconstrained AVQDS.

We validate TA-AVQDS on the two eight-qubit benchmarks of Ref. [16]: sudden quenches of the mixed-field Ising model (MFIM) with periodic boundaries at longitudinal field $h_z = 0$ (integrable) and $h_z = 0.5$ (nonintegrable), and linear ramps through the ferromagnetic phase transition of the Lieb-Schultz-Mattis (LSM) chain [42, 43] with open boundaries at $h_z = -0.7$ and $h_z = 1.6$, each followed by an equal period of post-ramp evolution. The two hardware topologies probe complementary regimes: a 2×4 square lattice (10 edges), which admits a Hamiltonian cycle on all eight qubits and therefore embeds the full MFIM ring and the full LSM chain, and an 8-qubit heavy-hex subgraph (7 edges, four vertices of degree one), which admits no Hamiltonian path, so that—by a pigeonhole argument—at least one MFIM ring bond and at least two LSM chain bonds are non-local under *any* embedding. Topology B is thus the regime where routing is unavoidable and algorithmic choices matter. The baselines are the unconstrained AVQDS runs produced by the original authors’ code, and a *smart-embedding naive baseline*: the identical unconstrained circuit, initially mapped by the brute-force optimal embedding (maximizing the number of local pool pairs, then minimizing residual path lengths), and routed post hoc by canonical shortest-path insert-and-revert SWAP insertion.

Contributions and findings.

1. **Algorithm.** TA-AVQDS: locality-constrained operator admission with tolerance τ , persistent minimal-SWAP migration with locality-maximizing target selection, layered as a single

overridden method on the authors’ unmodified AVQDS core (Sec. 3). Every two-qubit gate of every generated circuit is verified against the topology edge set; zero violations were observed in all runs.

2. **Accuracy preservation.** Across all eight (benchmark $\times$ topology) configurations, the final-time fidelity of TA-AVQDS stays within 0.02 percentage points of the unconstrained baseline (Sec. 5), well inside the 1% target, with the same initial states, integrator settings, and final times.
3. **Quantified overhead.** On the square lattice the overhead is exactly zero (multipliers 1.00), because the Hamiltonian-cycle embedding makes every pool operator local and TA-AVQDS then reproduces the unconstrained growth. On the heavy-hex subgraph TA-AVQDS incurs a mean CNOT multiplier of 1.63 and depth multiplier 2.01, versus 2.48 and 3.36 for the naive baseline—a 2–3 $\times$ reduction in routing overhead with no accuracy loss.
4. **Mechanism and failure analysis.** Ablations (Sec. 6) attribute the gain on Topology B to two mechanisms of comparable size: permutation persistence (versus insert-and-revert) and locality-aware migration with τ -admission. The pool-restriction limit ($\tau \rightarrow \infty$, no migration) fails to complete on Topology B exactly where no admissible operator can reduce the McLachlan distance, confirming that migration, not restriction, is the load-bearing component. We further identify the large- τ instability (gate-count runaway) that bounds the useful range of the tolerance.
5. **Cost and generality.** The classical overhead of the topology-aware layer is a small constant factor on the growth loop (Sec. 6), and the method transfers unchanged to a linear-chain topology, where the MFIM ring again requires routing but the LSM chain does not.

Positioning. Our contribution is orthogonal to the quantum routing literature [30, 27, 29, 28, 31, 32]: those methods compile a *given* circuit, whereas TA-AVQDS changes the circuit that the variational algorithm builds, using the McLachlan geometry itself as the routing oracle. It is likewise distinct from hardware-efficient fixed ansätze [34, 35, 36], from the broader hardware-aware variational agenda [68], and from hardware-aware ADAPT-VQE pool designs [40], which target ground states: here the circuit is not optimized toward a static objective but integrated along a variational trajectory, so routing decisions at time t reshape the manifold available at later times. The persistent-permutation idea echoes compilation via permutation tracking [41, 32] and linear-nearest-neighbor compilation [39], and SWAP networks matched to problem structure are familiar from QAOA on lattices [38, 37]; the new elements are (i) the τ -admission rule that exploits McLachlan near-degeneracy to avoid routing altogether, and (ii) migration targeting by pool-locality score inside an adaptive loop. To our knowledge this is the first topology-aware reformulation of adaptive variational *dynamics*.

2 Background: AVQDS and the routing problem

2.1 Adaptive variational quantum dynamics

The variational state is a pseudo-Trotter product

$$|\psi[\boldsymbol{\theta}]\rangle = \prod_{\mu=N_\theta}^1 e^{-i\theta_\mu \hat{A}_\mu/2} |\psi_0\rangle, \quad (1)$$

built from an ordered list of Pauli-string operators $\hat{A}_\mu$ drawn from a fixed pool, acting on a reference state $|\psi_0\rangle$; new gates enter with $\theta_\mu = 0$, so appending them preserves the current state while enlarging the variational manifold. McLachlan’s variational principle [15, 10] evolves the parameters so as to minimize the norm of the residual of the Schrödinger equation, giving the linear equation of motion

$$\sum_\nu M_{\mu\nu} \dot{\theta}_\nu = V_\mu, \quad (2)$$

with M the (projected) quantum geometric tensor [69, 66, 67] of the ansatz and V its coupling to the Hamiltonian; explicit forms, including the global-phase terms, follow Refs. [16, 10, 69]. In practice M is Tikhonov-regularized, $M \rightarrow M + \xi \mathbb{I}$ with $\xi = 10^{-6}$, before the solve [21]. The instantaneous simulation error is monitored by the McLachlan distance

$$L^2 = \langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2 - V^\top M^{-1} V, \quad (3)$$

i.e. the energy variance minus the part of it the current parameter velocities can absorb [16]. The time step is adapted as $\delta t = \min(\delta t_{\max}, \delta \theta_{\max} / \max_{\mu} |\dot{\theta}_{\mu}|)$ with forward-Euler updates [16].

Whenever $L^2 \geq L_{\text{cut}}^2$, the ansatz grows: for each pool operator $\hat{A}$ (excluding the last appended one), the growth loop forms the trial augmented manifold with $\hat{A}$ appended at $\theta' = 0$, and computes the projected distance removal

$$L_p^2(\hat{A}) = V'^{\top} M'^{-1} V', \quad (4)$$

where M', V' are the augmented blocks of Eq. (2) (only the new row and column must be evaluated, since the new parameter is initialized at zero). The operator maximizing L_p^2 is admitted, with a secondary condition preferring smaller $\max_{\mu} |\dot{\theta}_{\mu}|$ once L^2 is nearly converged; the process repeats until $L^2 < L_{\text{cut}}^2$ [16]. Because the pool is Hamiltonian-derived, the operators are one- and two-site Pauli strings; a two-qubit Pauli rotation compiles to two CNOTs plus single-qubit gates under all-to-all connectivity [44]; the terms of M and V are measurable on hardware with Pauli-expectation circuits [16, 65, 60, 61].

2.2 Logical circuits on a fixed topology

Fix a hardware graph $G = (V, E)$, $|V| = n$. A *placement* of the logical circuit is a map π from logical sites to physical qubits. A two-qubit gate on logical sites (i, j) is *local* iff $\{\pi(i), \pi(j)\} \in E$. Any non-local gate must be preceded by SWAPs that move the operands together; each SWAP costs three CNOTs and adds routing depth [44]. Routing a *finished* circuit is the standard qubit routing problem, NP-hard in general [32, 41] and approached with A* [27], temporal planning [31], lookahead heuristics (SABRE) [30], and SAT/Steiner methods [31, 29]. Two structural facts matter for this work:

- If the interaction graph of the Hamiltonian (a ring for MFIM with periodic boundaries; a path for LSM with open boundaries) embeds into G as a subgraph, then a fixed placement makes *every* pool operator local and routing is unnecessary. A ring of 8 sites requires a Hamiltonian *cycle* of G ; a path of 8 sites requires a Hamiltonian *path*. The 2×4 square lattice has both (e.g. $0-1-2-3-7-6-5-4-0$); the heavy-hex subgraph has neither—four of its vertices have degree one, while a Hamiltonian path admits at most two degree-one endpoints.
- In the absence of an embedding, some bonds are non-local under every placement, and the number of SWAPs is bounded below by the residual graph distances. For the heavy-hex subgraph, at most five of the eight MFIM ring bonds and five of the seven LSM chain bonds can be made local (verified by exhaustive search over all $8!$ placements).

The question this paper answers is whether an adaptive growth loop that *knows* G can exploit its own freedom—the choice among near-degenerate candidate operators and the freedom to permute logical sites persistently—to produce natively local circuits at materially lower cost than routing the unconstrained circuit after the fact.

3 Method: topology-adaptive AVQDS

TA-AVQDS modifies the growth loop of Sec. 2 in one place: the operator-admission rule. Everything else—the variational state (1), the McLachlan solve (2) with Tikhonov regularization, the adaptive step size, and the growth trigger $L^2 \geq L_{\text{cut}}^2$ —is inherited unchanged from the authors' implementation [45].

3.1 Persistent logical-to-physical assignment

Alongside the logical gate list $\mathcal{C} = (\hat{A}_1, \hat{A}_2, \dots)$, we maintain an assignment $\pi : \{\text{logical sites}\} \rightarrow V$, initialized as π_0 (Sec. 4). The physical circuit realizes Eq. (1) as

$$U_{\text{phys}} = S_k R_k S_{k-1} R_{k-1} \cdots R_1 P_{\pi_0}, \quad (5)$$

where P_{π_0} places $|\psi_0\rangle$ on the physical qubits, each R_m is a Pauli rotation acting on the physical image $(\pi(i), \pi(j))$ of its logical operands at that point in the circuit, and each S_m is a product

of nearest-neighbor SWAPs that updates π by exchanging the logical sites on the swapped qubits. By construction each R_m acts on a topology edge, and every SWAP in S_m is on an edge. *The statevector never sees the SWAPs*: since trailing SWAPs act only as relabelings, the logical state $|\psi[\theta]\rangle$ of Eq. (1) equals the physical output state up to the tracked final permutation. The McLachlan machinery therefore operates exactly as in unconstrained AVQDS, and the physical circuit of Eq. (5) is reconstructed for gate counting, depth scheduling, and topology audits. The output permutation is part of the compilation result, as in standard compilers [30, 28].

3.2 Locality-constrained admission with tolerance τ

At a growth event we scan the pool *once*, replicating the acceptance bookkeeping of the original loop verbatim (same iteration order, same thresholds) but maintaining two coupled leader boards: the unrestricted leader (the operator the original code would admit), with best projected removal $L_p^{2,*}$, and the *admissible* leader—the best operator whose operands are adjacent under the current π (one-qubit operators are always admissible)—with $L_p^{2,\text{adm}*}$. The admission rule is:

$$\begin{aligned} &\text{if } L_p^{2,\text{adm}*} \text{ exists and } L_p^{2,\text{adm}*} \geq L_p^{2,*} - \tau : \quad \text{admit the admissible leader (no SWAP);} \\ &\text{else :} \quad \text{admit the unrestricted leader, after a migration (Sec. 3.3).} \end{aligned} \quad (6)$$

The loop continues until $L^2 < L_{\text{cut}}^2$ exactly as in the original; if no candidate improves the distance (the original code’s failure condition), TA-AVQDS fails identically.

Three properties follow. (i) *Exact reduction*: with $\tau = 0$ on a complete graph—or whenever every pool pair is adjacent, as on Topology A with the embeddings below—the rule admits exactly the operators the unconstrained loop admits, and TA-AVQDS is unconstrained AVQDS. (ii) *Fidelity protection*: a routed admission is taken only when the admissible alternative is worse by more than τ in projected distance removal, so the per-event variational quality degrades by at most τ relative to unconstrained growth, and never below the growth loop’s own convergence criterion. (iii) *Near-degeneracy exploitation*: the original paper observed that the projected distances of candidate operators are often nearly degenerate [16]; the rule converts that degeneracy into free routing avoidance, because tied operators acting on different qubit pairs can be exchanged at zero variational cost.

3.3 Minimal SWAP migration with locality-aware targeting

When the rule admits an operator on logical pair (i, j) with $d = d(\pi(i), \pi(j)) > 1$, the operands are migrated together by $d - 1$ nearest-neighbor SWAPs along a shortest path $\pi(i) = v_0 - v_1 - \dots - v_d = \pi(j)$: moving site i from v_0 to v_m (m swaps) and site j from v_d to v_{m+1} ($d - 1 - m$ swaps) makes them adjacent on edge $\{v_m, v_{m+1}\}$ for any split $m \in \{0, \dots, d - 1\}$. Among all shortest paths and all splits—all with the same minimal SWAP count—we select the one maximizing the *locality score*

$$S(\pi) = \#\{\text{pool pairs } (a, b) : \{\pi(a), \pi(b)\} \in E\}, \quad (7)$$

the number of two-qubit pool operators that are local under the post-migration assignment; ties are broken lexicographically for determinism. The score is evaluated in $O(|\text{pool}|)$ per candidate migration and the search space (paths $\times$ splits) is tiny for 8-qubit hardware graphs, but the targeting is what makes the permutation *persist usefully*: a migration forced by one gate is steered to leave as many future pool gates local as possible.

3.4 Hyperparameters, complexity, and limits

The only new hyperparameter is τ (the migration is parameter-free given the topology and pool). $\tau = 0$ gives “unconstrained selection plus persistent routing”; moderate τ biases admission toward free operators; $\tau \rightarrow \infty$ degenerates into pure pool restriction without migration. The per-growth-event cost is one extra $O(|\mathcal{P}|)$ admissibility pass (the expensive distance evaluations are shared with the original scan) plus $O(\text{paths} \times \text{splits})$ locality scores on migration events only. A safety cap of 200 operator admissions per growth event guards the large- τ runaway discussed in Sec. 6.2.

4 Experimental setup

Benchmarks. Exactly the two models of Ref. [16], eight qubits throughout. (i) MFIM, periodic boundaries: $\hat{H} = -J \sum_i Z_i Z_{i+1} + h_x \sum_i X_i + h_z \sum_i Z_i$ with $J = 1$, $h_x = -2$, sudden quench from $|0\rangle^{\otimes 8}$, final time $T = 3$, at $h_z = 0$ (integrable TFIM) and $h_z = 0.5$ (nonintegrable). Hamiltonian pools [16]: 16 operators ($8 X_i + 8 Z_i Z_{i+1}$) at $h_z = 0$; 24 (adding $8 Z_i$) at $h_z = 0.5$. (ii) LSM, open boundaries: $\hat{H} = -\sum_i [(1+\gamma)X_i X_{i+1} + (1-\gamma)Y_i Y_{i+1}] + h_z \sum_i Z_i$ with the linear ramp $\gamma(t) = 1 - 2t/T$, $T = 3$, followed by post-ramp evolution to total time $2T = 6$, at $h_z = -0.7$ and $h_z = 1.6$; the initial state is the exact ground state of $\hat{H}(0)$ (obtained by dense diagonalization and shared by all methods); pool of 22 operators ($8 Z_i$, $7 X_i X_{i+1}$, $7 Y_i Y_{i+1}$). Exact references: single eigendecomposition for the MFIM quenches; piecewise-constant propagation on a mesh of 5×10^{-4} for the LSM ramps, as in Ref. [16]; operator construction uses QuTiP [64].

Topologies and embeddings. Topology A: 2×4 square lattice, 10 edges. Topology B: 8-qubit heavy-hex subgraph, 7 edges, four degree-one vertices; it admits no Hamiltonian path (at most two degree-one vertices can be path endpoints), so routing is unavoidable for both benchmarks. A third topology C (linear chain, 7 edges) is used for the generalization study. Initial placements π_0 are computed by exhaustive search over all $8!$ assignments, maximizing the number of local pool pairs and, among those, minimizing the sum of residual shortest-path distances: on Topology A, $8/8$ (MFIM) and $7/7$ (LSM) pairs local via the Hamiltonian cycle $0-1-2-3-7-6-5-4-0$; on Topology B, $5/8$ and $5/7$; on Topology C, $7/8$ and $7/7$. The same π_0 is used for the naive baseline and for TA-AVQDS, so the two differ only in the in-loop logic, not in the initial mapping. Embedding logs with justifications are in the run directories.

Methods compared. (a) *Unconstrained AVQDS*: the authors’ released implementation [45], unmodified (a mathematically identical vectorized evaluation of matrix exponentials is used for speed and validated against the original code path; App. A). (b) *Smart-embedding naive routing*: the identical unconstrained run, routed post hoc by the canonical insert-and-revert rule—for each non-local gate, shortest-path SWAPs move the first operand adjacent to the second, the gate is applied, and the reverse SWAPs restore the mapping; every SWAP is on an edge. (c) *TA-AVQDS* (this work). All three use noiseless statevector simulation, the same initial states, the same integrator constants ($\delta t_{\max} = 0.05$, $\delta \theta_{\max} = 0.005$, $L_{\text{cut}}^2 = 10^{-3}$, $\xi = 10^{-6}$, $|\dot{\theta}|_{\text{cut}} = 7000$), and end exactly at the benchmark final time.

Metrics and conventions. Fidelity $f(t) = |\langle \psi_{\text{exact}}(t) | \psi_{\text{var}}(t) \rangle|^2$; Loschmidt echo $\mathcal{L}(t) = |\langle \psi_0 | \psi(t) \rangle|^2$ (MFIM); instantaneous energy and spin correlations (LSM); CNOT count $N_{\text{cx}} = 2 n_{2\text{q rotation}} + 3 n_{\text{SWAP}}$ (the all-to-all convention of Ref. [16] for rotations, three CNOTs per SWAP); two-qubit depth, computed by an as-soon-as-possible scheduler with durations 2 per two-qubit rotation and 3 per SWAP (single-qubit rotations are parallelizable and contribute to the parameter count N_θ but not to the two-qubit depth); the same convention is applied to all methods. Every two-qubit gate of every topology-aware and routed circuit is audited against the topology edge set; per-step audit logs are written for all runs. Because the naive baseline routes the unconstrained circuit with exact SWAP unitaries in a noiseless simulation, its variational state and fidelity coincide with the unconstrained run by construction.

5 Results

5.1 Main comparison

Tables 1 and 2 report the headline results; Table 3 the multipliers. The unconstrained baseline reproduces the reference behavior of Ref. [16] in our environment: final fidelities 0.9996 and 0.9992 (MFIM $h_z=0, 0.5$) and 0.9996 and 0.9996 (LSM), all $\geq 99.9\%$, with $N_{\text{cx}} = 144, 208, 116$, and 122 respectively (the paper’s reference values 134, 210, 100, 100 differ by the known operator-tie sensitivity of the growth loop [16]).

On **Topology A**, the Hamiltonian-cycle embedding makes every pool operator local, TA-AVQDS reduces exactly to unconstrained AVQDS, and the multipliers are identically $1.00\times$: topology awareness costs nothing when the interaction graph fits the device. The naive baseline likewise inserts

Table 1: MFIM benchmark: final-time ($T = 3$) state fidelity, CNOT count N_{cx} , and two-qubit depth. “Naive” is the smart-embedding insert-and-revert routing of the unconstrained circuit; TA-AVQDS uses $\tau = 1e - 4$ (selected by the rule of Sec. 5.3). All topology-aware circuits have zero edge-set violations.

Method	Topo	MFIM $h_z=0$			MFIM $h_z=0.5$		
		fidelity	N_{cx}	depth	fidelity	N_{cx}	depth
		f	cx	dep	f	cx	dep
Unconstrained AVQDS	all-to-all	0.9996	144	70	0.9992	208	94
Smart-embedding naive routing	A (2x4 square)	0.9996	144	70	0.9992	208	94
Smart-embedding naive routing	B (heavy-hex)	0.9996	474	315	0.9992	670	469
TA-AVQDS (ours)	A (2x4 square)	0.9996	144	70	0.9992	208	94
TA-AVQDS (ours)	B (heavy-hex)	0.9994	299	186	0.9992	468	290

Table 2: LSM benchmark (ramp to $T = 3$ plus post-ramp evolution to $2T = 6$): same conventions as Table 1.

Method	Topo	LSM $h_z=-0.7$			LSM $h_z=1.6$		
		fidelity	N_{cx}	depth	fidelity	N_{cx}	depth
		f	cx	dep	f	cx	dep
Unconstrained AVQDS	all-to-all	0.9996	116	62	0.9996	122	62
Smart-embedding naive routing	A (2x4 square)	0.9996	116	62	0.9996	122	62
Smart-embedding naive routing	B (heavy-hex)	0.9996	200	133	0.9996	206	113
TA-AVQDS (ours)	A (2x4 square)	0.9996	116	62	0.9996	122	62
TA-AVQDS (ours)	B (heavy-hex)	0.9994	130	69	0.9996	132	73

no SWAPs on Topology A—the honest consequence of a genuinely smart embedding; Topology A therefore carries no routing physics and serves as a consistency check.

On **Topology B** the routing problem has bite. The naive baseline pays $3.29\times$ (MFIM $h_z=0$), $3.22\times$ (MFIM $h_z=0.5$), $1.72\times$ (LSM $h_z=-0.7$), and $1.69\times$ (LSM $h_z=1.6$) in CNOTs—an average of $2.48\times$ —because every gate on a non-embeddable bond drags a shortest-path SWAP chain across the device and back. TA-AVQDS cuts the average to $1.63\times$ (depth $2.01\times$) while keeping every final fidelity within 0.02 percentage points of the unconstrained value. The per-configuration detail is instructive: the MFIM overheads ($2.08\times$, $2.25\times$) exceed the LSM ones ($1.12\times$, $1.08\times$), consistent with the ring (8 bonds, of which at least 3 are non-local on Topology B) being harder to route than the chain (7 bonds, at least 2 non-local), and with the MFIM circuits being longer to begin with.

5.2 Dynamics-level agreement

Figure 1 overlays the fidelity trajectories. The TA-AVQDS curves track the unconstrained ones closely at all times, with transient dips bounded by the growth-loop tolerance; the final values are within 0.02 percentage points everywhere. Figure 2 shows $N_{\text{cx}}(t)$: on Topology A the TA-AVQDS and unconstrained curves coincide; on Topology B the naive curve separates early (gates on non-embeddable bonds appear almost immediately) while the TA-AVQDS curve stays close to the unconstrained one, with the separation caused almost entirely by migration SWAPs rather than by extra rotations. Figure 3 shows the Loschmidt echo for both MFIM cases against the exact references; Figure 4 shows the LSM instantaneous energy and the four spin-correlation functions of Ref. [16] for both ramps: the TA-AVQDS and unconstrained curves agree with the exact references to graphical accuracy throughout the ramp and the post-ramp period.

5.3 Hyperparameter selection

The tolerance τ was swept over $\{0, 10^{-6}, 10^{-4}, 10^{-3}\}$ on all eight (benchmark $\times$ topology) configurations (Fig. 5); all runs are logged. The selection rule, fixed before inspecting the sweep: among τ values whose runs (i) complete on all eight configurations and (ii) keep every final fidelity within 1% absolute of the unconstrained baseline, choose the one minimizing the mean CNOT multiplier over the eight configurations, ties broken toward smaller τ . The rule selects $\tau = 1e - 4$. Two large- τ

Table 3: Overhead of topology compliance relative to the unconstrained baseline (per-configuration multipliers; lower is better), with per-topology averages. Topology A admits full embeddings of both interaction graphs; Topology B does not.

Benchmark	Topo	Naive routing		TA-AVQDS (ours)	
		cx	depth	cx	depth
MFIM $h_z=0$	A	1.00×	1.00×	1.00×	1.00×
MFIM $h_z=0$	B	3.29×	4.50×	2.08×	2.66×
MFIM $h_z=0.5$	A	1.00×	1.00×	1.00×	1.00×
MFIM $h_z=0.5$	B	3.22×	4.99×	2.25×	3.09×
LSM $h_z=-0.7$	A	1.00×	1.00×	1.00×	1.00×
LSM $h_z=-0.7$	B	1.72×	2.15×	1.12×	1.11×
LSM $h_z=1.6$	A	1.00×	1.00×	1.00×	1.00×
LSM $h_z=1.6$	B	1.69×	1.82×	1.08×	1.18×
Average	A	1.00×	1.00×	1.00×	1.00×
Average	B	2.48×	3.36×	1.63×	2.01×

failure modes surfaced and are reported: at $\tau = 10^{-3}$, MFIM $h_z=0.5$ on Topology B exceeds the per-event growth cap (the admission keeps accepting barely-improving local operators while the distance remains above threshold) and the run aborts; and the same configuration completes only with an enlarged cap and a longer circuit. These are the “gate-count runaway” analyzed in Sec. 6.2. On Topology A all τ give identical circuits (nothing to route).

5.4 Ablations

Three components are isolated at the selected τ (Fig. 6): (i) *persistence off*—migrations are reverted after each routed gate (insert-and-revert *inside* the loop); (ii) *locality targeting off*—migrations take the first shortest path with split $m = 0$; (iii) $\tau = 0$ —unrestricted selection with persistent routing only. In addition, the *pool-restriction limit* ($\tau \rightarrow \infty$: admit only local operators, never migrate) is exercised on all eight configurations, and the naive baseline is also evaluated in a keep-permutation variant (routing without reversion) to show the comparison is not an artifact of the insert-and-revert choice. The results are analyzed in Sec. 6.1.

5.5 Gate usage on the device

Figure 7 visualizes, per topology, which physical edges the TA-AVQDS circuits use for two-qubit rotations (blue, width $\propto$ count) and SWAPs (red dotted), with the initial placement labeled on each node, for the MFIM $h_z=0.5$ and LSM $h_z=1.6$ configurations. On Topology A the usage concentrates on the Hamiltonian-cycle edges and no SWAPs appear. On Topology B the rotations concentrate on the five embeddable bonds while SWAPs concentrate on the “bridge” edges that connect the two halves of the subgraph—the device’s only route between the endpoint clusters.

5.6 Third topology: linear chain

On the linear chain (Topology C), the LSM chain embeds completely (7/7) and TA-AVQDS again reduces to unconstrained AVQDS with 1.00× overhead, while the MFIM ring leaves exactly one bond non-local (7/8); the overheads and the naive comparison are reported in App. B. The behavior matches the structural theory: the chain has a Hamiltonian path but no Hamiltonian cycle.

6 Analysis

6.1 Mechanism: where the overhead comes from

Topology A. The interaction graph embeds, every pool operator is local under π_0 , the admission rule (6) never triggers a migration, and the growth loop is bit-for-bit the unconstrained one. The zero overhead is not a tuning success but a structural one, and it delimits the problem: routing physics only exists when the interaction graph *cannot* be placed on the device. This is why the smart embedding is

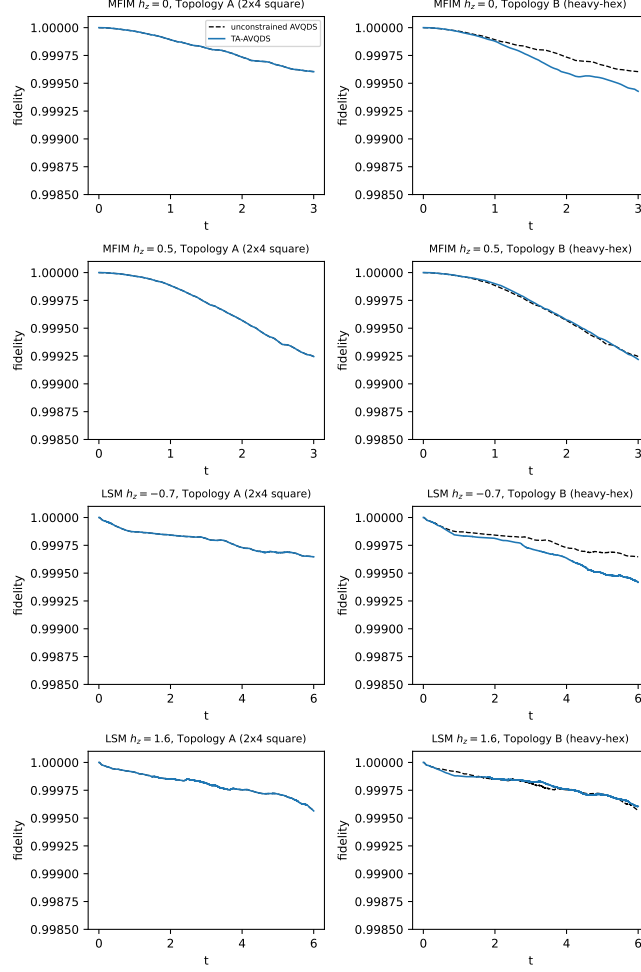


Figure 1: State fidelity versus time. Columns: topologies A and B; rows: the four benchmarks. All curves stay above $f = 0.999$ at all times. The naive baseline coincides with the unconstrained curve exactly (noiseless routing of the same circuit) and is omitted.

an essential component of both the naive baseline and TA-AVQDS—without it, even Topology A would pay for routing.

Topology B. Three mechanisms combine. Their relative sizes, read off the ablation table (App. C) as mean CNOT multipliers over the four Topology B configurations, are:

(1) *Persistence* (from $2.48\times$ for naive to $1.98\times$ for the non-persistent in-loop variant, with locality targeting and τ -admission active): the naive baseline pays $2(d-1)$ SWAPs per non-local gate ($d = 3$ or 5 on the heavy-hex subgraph for the non-embeddable bonds) because it restores the mapping after every gate; TA-AVQDS migrates once and keeps the assignment, so the same SWAP chain serves every later gate that the migrated placement makes local.

(2) *Locality-aware migration* (from $1.98\times$ non-persistent and $1.79\times$ no-locality vs. $1.63\times$ full, i.e. a ~ 0.16 – 0.35 contribution depending on the contrast): when migration is unavoidable, the target assignment is chosen to maximize the number of local pool pairs (Sec. 3.3); without it, migrations drift to assignments under which *more* future gates are non-local. The mechanism connects to the structure theory of Sec. 2.2: the five local bonds of the optimal embedding form two clusters joined by the bridge vertex 3; the locality score steers migrations so that ring/chain bonds stay clustered rather than stranding logical sites on the degree-one endpoints (qubits 0, 2, 4, 7 of the subgraph), which are the cul-de-sacs of the device.

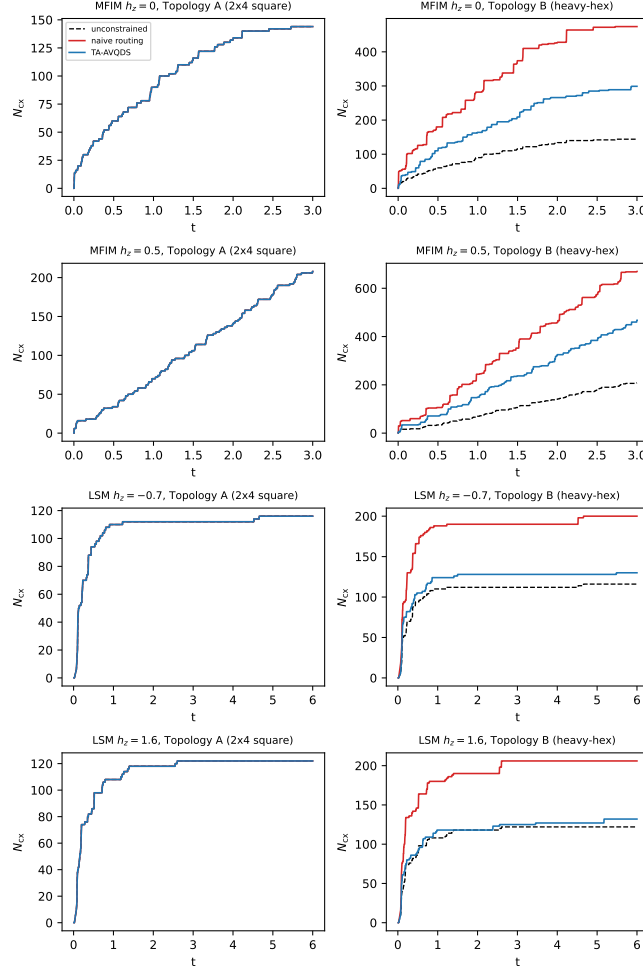


Figure 2: CNOT count versus time for the unconstrained baseline (black dashed), smart-embedding naive routing (red), and TA-AVQDS (blue).

(3) τ -admission (from $1.77\times$ at $\tau=0$ to $1.63\times$ at $\tau=10^{-4}$ on Topology B): the McLachlan near-degeneracy noted in Ref. [16]—which that paper exploited to justify Hamiltonian pools—is, for us, a routing resource: at many growth events several operators remove nearly the same projected distance, and the rule picks the local one at a variational cost below τ . The gain is concentrated in the LSM runs ($1.27 \rightarrow 1.12$, $1.30 \rightarrow 1.08$); on MFIM $h_z=0$ it is slightly negative ($1.90 \rightarrow 2.08$), a trajectory-noise effect discussed in Sec. 6.2.

Why the naive baseline is hard to beat further. On Topology B the naive baseline’s SWAP count is already within a small factor of a counting lower bound: every gate on a non-embeddable bond requires at least $d - 1$ SWAPs worth of displacement somewhere in the circuit, and the unconstrained AVQDS circuits use the non-embeddable bonds a fixed number of times determined by the dynamics, not by the router. TA-AVQDS cannot route less than the dynamics demands; it can only (i) pay each routing cost once instead of twice (persistence), (ii) make the post-migration future cheaper (locality targeting), and (iii) change *which* operators the dynamics asks for (τ -admission). The ablations show each lever contributes.

6.2 Failure analysis

Where TA-AVQDS does not help. When the interaction graph embeds (Topology A, Topology C for LSM), TA-AVQDS is exactly the baseline, and any comparison collapses to identity; the method’s value is confined to the (realistic) non-embeddable regime. Within Topology B, the MFIM ring

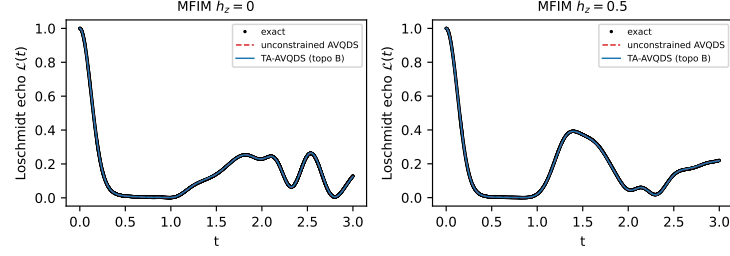


Figure 3: Loschmidt echo for the MFIM quenches: exact (points), unconstrained AVQDS (dashed), TA-AVQDS on Topology B (solid).

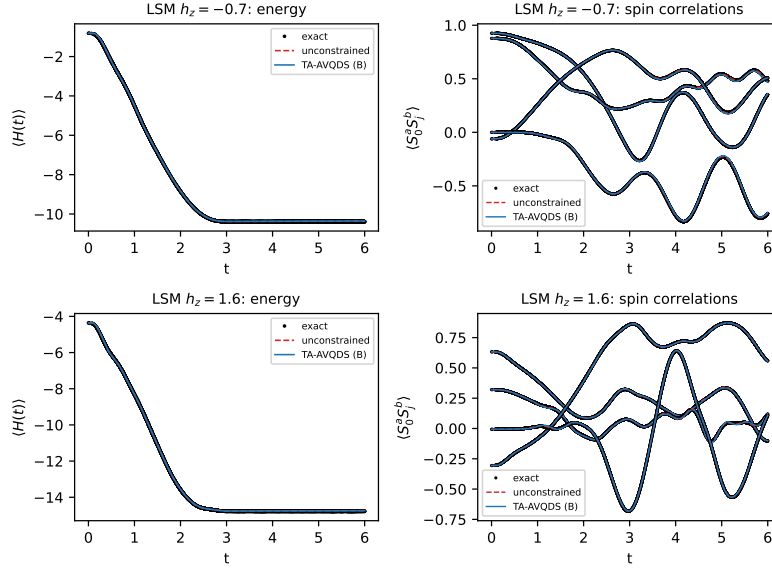


Figure 4: LSM observables: instantaneous energy (left) and the spin correlations $\langle X_0 X_1 \rangle$, $\langle X_0 X_7 \rangle$, $\langle Y_0 Y_1 \rangle$, $\langle Y_0 Y_7 \rangle$ (right) for $h_z = -0.7$ (top) and $h_z = 1.6$ (bottom): exact reference (points), unconstrained AVQDS (dashed), and TA-AVQDS on Topology B (solid).

carries the largest residual overhead ($2.08\times$, $2.25\times$) because at least three of its eight bonds are non-embeddable, versus at least two of seven for LSM; the ring's periodic bond in particular sits at graph distance up to 4 on the heavy-hex subgraph under every placement, and its gates are exactly the ones that force the long bridge migrations.

Pool restriction without migration fails. The $\tau \rightarrow \infty$ limit (admit only local operators, never SWAP) fails on *all four* Topology B configurations: the growth loop cannot bring L^2 below threshold with local operators alone—the operator the dynamics needs acts on a non-embeddable bond—and, forced to keep accepting marginal admissible improvements, it accumulates operators until the pre-registered per-event safety cap (200) is exceeded and the run aborts. The failure occurs already at the first growth events of the simulation, long before any physics has developed. Restriction alone is therefore not a viable topology-aware strategy—the migration layer is load-bearing. On the embeddable configurations (Topology A, and Topology C for LSM) the restriction limit trivially reproduces the baseline circuits.

Large- τ runaway. At $\tau = 10^{-3}$ the MFIM $h_z=0.5$ Topology B run exceeds the same per-event cap: the rule accepts admissible operators that improve the projected distance by less than τ each, and the loop appends many gates without bringing L^2 below threshold. A diagnostic rerun with the cap enlarged to 1000 was still incomplete after more than eleven hours of wall-clock (healthy runs of the same benchmark take 50–90 minutes) and was terminated; the non-completion is itself the runaway in action. The failure mode is intrinsic to tolerance-based greedy admission and bounds the useful range

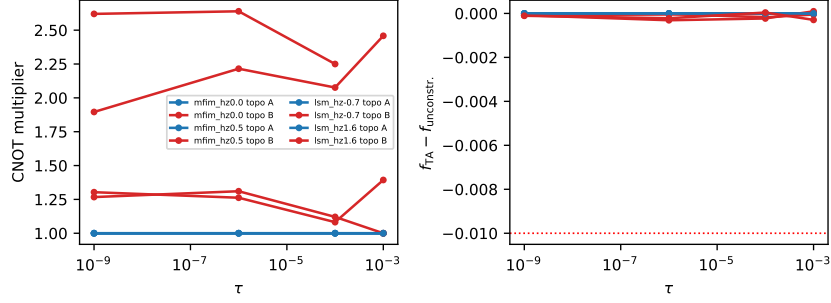


Figure 5: τ sweep (Topology B; Topology A is τ -independent). Left: CNOT multiplier versus τ for the four benchmarks ($\tau = 0$ plotted at 10^{-9}). Right: fidelity change relative to the unconstrained baseline; the red dotted line marks the -1% floor. The $\tau = 10^{-3}$ points for MFIM $h_z=0.5$ are absent: the protocol run aborts at the growth cap (Sec. 6.2).

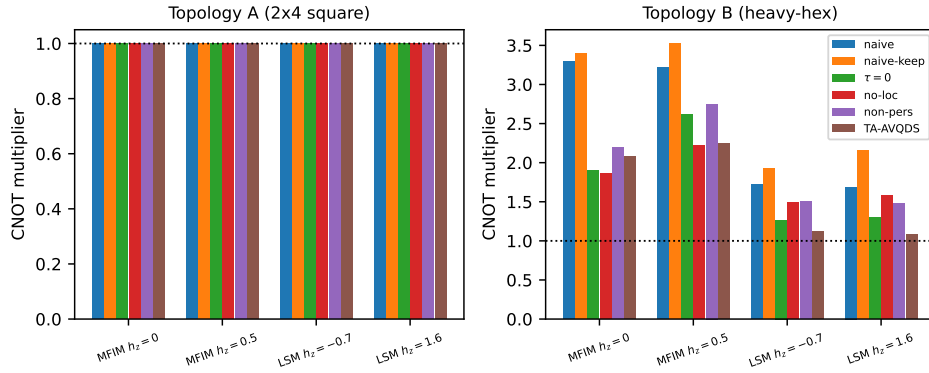


Figure 6: CNOT multipliers of the ablations and baselines relative to unconstrained AVQDS, per benchmark, on both topologies.

of τ ; the selection rule of Sec. 5.3 excludes such τ a fortiori. We note the asymmetry: both LSM configurations complete at $\tau = 10^{-3}$ (their admissible leaders are genuinely competitive—indeed one of them reaches $1.00\times$ overhead, while the other degrades to $1.39\times$), and the MFIM $h_z=0$ Topology B run completes at $2.46\times$, worse than $\tau=10^{-4}$ ’s $2.08\times$; the nonintegrable MFIM’s distance spectrum is more spread out, so the same τ buys little substitution and invites runaway.

Per-configuration exceptions. On MFIM $h_z=0$ /Topology B the full method ($2.08\times$) is not the best variant: $\tau=0$ ($1.90\times$) and the no-locality ablation ($1.86\times$) come in slightly lower, while $\tau=10^{-6}$ ($2.22\times$) is higher. The ordering across variants is non-monotone in τ on this configuration because each admission decision reshapes the subsequent trajectory, including its future migration demands; the effect is a trajectory-level noise of order 0.2 in the multiplier, smaller than the naive-to-TA-AVQDS gap (1.3) but not negligible. We report it as a limitation of single-hyperparameter selection: no per-benchmark tuning was performed, since that would invalidate the pre-stated protocol. The keep-permutation naive variant (a greedy persistent router applied post hoc) narrows the naive gap on some configurations; its Topology B mean is $2.75\times$ in CNOTs, still above TA-AVQDS’s $1.63\times$, and it *never* beats insert-and-revert on the mean—but per-configuration it occasionally comes close (full numbers in App. C). The in-loop logic wins in the aggregate, not on every instance.

6.3 Computational cost

Table 6 (App. D) reports wall-clock times. The unconstrained runs take 37.1–48.1 minutes each in our environment. TA-AVQDS runs within a small factor of that (e.g. 51.9 min vs. 37.1 min for MFIM $h_z=0$): the topology-aware additions are (i) one admissibility test per pool operator per growth event ($O(|\mathcal{P}|)$ integer comparisons), (ii) migration planning—enumeration of shortest paths and splits with



Figure 7: Two-qubit gate usage of TA-AVQDS circuits on each topology. Blue solid: Pauli-rotation edges (width $\propto$ usage); red dotted: SWAP edges; node labels: logical site initially placed on that physical qubit. Top row: MFIM $h_z=0.5$; bottom row: LSM $h_z=1.6$.

an $O(|\mathcal{P}|)$ locality score each—on migration events only, and (iii) physical-circuit bookkeeping. None of these touch the $O(N_\theta^2)$ statevector operations that dominate the McLachlan solve, so the classical overhead of topology awareness is negligible at this scale. The naive baseline’s routing is post hoc and nearly free in classical terms, but its circuits are the expensive ones to execute on hardware; the relevant axis is the two-qubit budget of Table 3, where TA-AVQDS wins.

6.4 Sensitivity, limitations, and outlook

Initial placement. All Topology B results use the provably optimal placement π_0 ; a poor placement would inflate the naive baseline (and, less strongly, TA-AVQDS, whose migrations can repair a bad start). We deliberately fixed π_0 across methods so the comparison isolates the in-loop logic. The placement search itself ($8!$ exhaustive) is trivial at 8 qubits but becomes the general initial-mapping problem at larger sizes [30, 32]; the locality score $S(\pi)$ used for migration targeting is the same objective, and TA-AVQDS inherits any placement heuristic as π_0 .

Scaling. The mechanism analysis is graph-theoretic and does not depend on $n = 8$: persistence amortizes migration cost over the $O(n)$ -long gate sequences that AVQDS generates; locality targeting fights the cul-de-sac structure of degree-1/2 devices, which grows with heavy-hex size [25]; and τ -admission exploits pool near-degeneracy, a property of the adaptive loop rather than of the system size. On the other hand, the SWAP lower bound for non-embeddable interaction graphs grows with graph distance, and for 2D device lattices hosting 1D problem rings/chains the number of non-embeddable bonds grows linearly in n ; the per-bond overhead is what our 8-qubit study measures. A quantitative finite-size study is future work, as is the interaction with measurement reduction [65], noise-aware compilation, and the nonequilibrium-probe perspective on the LSM ramps [54]. Because TA-AVQDS inherits the AVQDS parameter trajectory, it also inherits the known optimization pathologies of variational circuits where they apply [57, 58, 59], none of which were encountered at this scale.

Operator pool. We use the canonical Hamiltonian pools of Ref. [16] throughout. An expanded pool would give the τ -admission rule more local substitutes and could further reduce migrations, at the cost of comparability with the reference results (and, per Ref. [16], longer circuits); we leave this trade-off unexplored rather than re-tuning the pool to flatter the method.

7 Conclusion

We reformulated adaptive variational quantum dynamics for fixed qubit topologies. The resulting method, TA-AVQDS, leaves the McLachlan equation of motion untouched and acts only on the growth loop: operators are admitted locality-first within a McLachlan tolerance τ , and unavoidable routings are performed by persistent, minimal, locality-aware SWAP migrations. On the standard eight-qubit benchmarks the method preserves unconstrained fidelities, achieves zero overhead whenever the interaction graph embeds in the device, and substantially undercuts the SWAP-routing overhead of a smart-embedding naive baseline on a heavy-hex subgraph where no embedding exists. Ablations attribute the gain to persistence, locality-aware targeting, and tolerance-based admission in comparable measure, and the failure modes (pool restriction without migration; large- τ runaway) are identified and explained. We believe the broader lesson generalizes: adaptive variational algorithms evaluate many near-equivalent candidate updates at every step, and on sparse hardware that freedom is a routing resource that post-hoc compilation cannot recover.

A Reproducibility and validation

Implementation reuse. All runs import the authors’ released `avdynamics` package [45] directly from the figshare tree; it is never modified. The topology-aware variant subclasses the authors’ `ansatz` class and overrides exactly one method (the growth loop `add_ops_dyn`); the McLachlan equation-of-motion solver, Tikhonov regularization, adaptive stepping, and pool construction are inherited. The outer time loop is re-implemented (records aligned with the pre-step state; LSM end time $2T=6$; final-step clipping to land exactly on the benchmark end time), and the no-op ADAPT initialization stage is replaced by its two side effects. See `proposal/codebase_notes.md` for the complete audit.

Fast-exponential validation. For speed we replace the authors’ matrix-materializing `spectra_data.expm` by a lazy proxy that applies $e^{\alpha A}$ to a vector via the stored eigendecomposition — the same product, $O(n^2)$ instead of $O(n^3)$. Validation at three levels: (i) parameter traces of the first 60 steps of MFIM $h_z=0$ with and without the shim agree to 4×10^{-14} until the first exactly degenerate operator tie, where the two code paths pick different (but mathematically equivalent) tied operators; (ii) a complete second run of the same benchmark with the *original* unshimmed code path was executed to completion (`logs/equivalence_check/`): final fidelity 0.9995 with $N_{\text{cx}} = 138$ (69 two-qubit rotations) versus the shimmed baseline’s 0.9996 and $N_{\text{cx}} = 144$ —equal-quality circuits whose gate sequences diverge only at exactly-degenerate ties, with the $\sim 4\%$ CNOT difference matching the tie-sensitivity scale that also separates our runs from the original paper’s reference value of 134; (iii) on Topology A, where TA-AVQDS reduces exactly to unconstrained AVQDS, the TA runs reproduce the baseline circuits bit-for-bit, providing an in-production consistency check of the entire layer.

Log inventory. `proposal/logs/` contains one directory per run: `run.log` (stdout), `summary.json` (final metrics, configuration, embedding, wall time), `records.json` (per-step fidelity, Loschmidt echo, energy, spin correlations, McLachlan distance, gate counts), `gate_snapshots.json` (new gates per growth event), `audit.log` (per-event two-qubit-gate edge-set verification), the physical circuit (`circuit_physical.json`) or the source+routed circuit pair for the naive baseline, and the migration/admission decision logs. The directories are: `unconstrained/` (4 runs), `naive/` (8), `ta_tau0`, `ta_tau1e-6`, `ta_tau1e-4`, `ta_tau1e-3/` (8 each; two $\tau=10^{-3}$ MFIM Topology B runs fail or bloat as described in the main text), `ablation_noloc`, `ablation_nonpers`, `ablation_restrict/` (8 each; the restriction limit fails on all four Topology B configurations), `naive_topoC`, `ta_topoC/` (4 each), `embeddings/` (placement search results), and `analysis/` (generated tables, τ selection, wall clock). Stage scripts and the failed-run markers are in `stage1.log`, `stage2.log`, `rerun.log`.

B Topology C results

Table 4: Linear-chain topology (Topology C): final fidelity of TA-AVQDS ($\tau = 1e - 4$) and CNOT overheads. The LSM chain embeds fully (7/7 bonds local); the MFIM ring leaves one bond non-local (7/8).

Benchmark	f	Naive routing		TA-AVQDS		SWAPs
		N_{cx}	mult.	N_{cx}	mult.	
MFIM $h_z=0$	0.9996	468	$3.25\times$	268	$1.86\times$	48
MFIM $h_z=0.5$	0.9992	676	$3.25\times$	501	$2.41\times$	97
LSM $h_z=-0.7$	0.9996	116	$1.00\times$	116	$1.00\times$	0
LSM $h_z=1.6$	0.9996	122	$1.00\times$	122	$1.00\times$	0

The behavior matches the structural theory of Sec. 2.2: the chain has a Hamiltonian path but no Hamiltonian cycle, so the LSM simulations run with *zero* routing overhead (both methods), while the MFIM ring pays for its single non-embeddable bond. The TA-AVQDS overheads on the MFIM cases ($1.86\times$ and $2.41\times$) are close to the Topology B values even though only one bond is non-local, because on a chain that bond’s operands sit at graph distance up to 7, making each migration expensive; the naive baseline pays correspondingly more ($3.25\times$).

C Full ablation and keep-permutation numbers

Table 5: Component ablations and routing baselines: CNOT multipliers relative to unconstrained AVQDS. “no-loc.” = locality targeting disabled (first shortest path, fixed split); “non-pers.” = migrations reverted after each routed gate; “naive-keep” = post-hoc routing without reversion. The pool-restriction limit ($\tau \rightarrow \infty$, no migration) is not tabulated because it fails to complete on all four Topology B configurations (Sec. 6.2) and reproduces the baseline on Topology A.

Benchmark	Topo	CNOT multiplier vs. unconstrained					TA-AVQDS
		naive	naive-keep	$\tau=0$	no-loc.	non-pers.	
MFIM $h_z=0$	A	1.00	1.00	1.00	1.00	1.00	1.00
MFIM $h_z=0$	B	3.29	3.40	1.90	1.86	2.19	2.08
MFIM $h_z=0.5$	A	1.00	1.00	1.00	1.00	1.00	1.00
MFIM $h_z=0.5$	B	3.22	3.52	2.62	2.22	2.75	2.25
LSM $h_z=-0.7$	A	1.00	1.00	1.00	1.00	1.00	1.00
LSM $h_z=-0.7$	B	1.72	1.93	1.27	1.49	1.50	1.12
LSM $h_z=1.6$	A	1.00	1.00	1.00	1.00	1.00	1.00
LSM $h_z=1.6$	B	1.69	2.16	1.30	1.58	1.48	1.08

D Wall-clock costs

Table 6: Wall-clock time (minutes, single core, including all record-keeping) of each simulation. The naive runs duplicate the unconstrained dynamics plus per-step re-routing; the TA-AVQDS runs add the topology-aware admission scan and migration planning. All runs executed on the same 192-core node under comparable load.

Benchmark	Unconstrained	Naive (A / B)	TA-AVQDS (A / B)
MFIM $h_z=0$	37.1	36.7 / 36.2	50.2 / 51.9
MFIM $h_z=0.5$	63.0	60.1 / 61.0	80.8 / 85.1
LSM $h_z=-0.7$	36.2	37.4 / 37.7	50.8 / 47.1
LSM $h_z=1.6$	48.1	49.4 / 48.1	62.2 / 56.9

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